

Experimental measurement and numerical simulation of an impinging jet with application to thunderstorm microburst winds

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Received 21 December 2005; received in revised form 20 June 2007; accepted 6 September 2007
Available online 7 November 2007

Abstract

Wind profiles and characteristics in a thunderstorm downburst are significantly different from that of regular boundary layer winds. This paper deals with the experimental and numerical simulation of a type of thunderstorm wind, namely the microburst, to study the outflow velocity characteristics. The microburst is simulated as a round jet, impinging onto a flat plate. A generic empirical equation for radial velocity profile is developed based on the experimental data, using hotwire, pressure rakes and particle image velocimetry (PIV). The experimental results are used to validate CFD simulations and to find the applicability of different turbulence models for this kind of flow. Favorable agreement between numerical and experimental studies indicates that CFD can be used for this kind of complex flow.

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Keywords: Microburst simulation; Impinging jet; RANS turbulence models; LES; PIV measurements; Velocity profile; Boundary-layer growth; Microburst wind loads

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1. Introduction

The current design wind loads for buildings and other structures are based on model tests in low-speed boundary-layer wind tunnels that generate straight-line winds. However, winds resulting from regular storm events such as thunderstorms are far from being regular atmospheric boundary-layer-type events. Thunderstorm winds have significant vertical velocity components and mean horizontal velocity distributions that are different from usual boundary-layer winds (as shown in Fig. 1). It is also believed that the gust structure in a downdraft is much better correlated over its width than in more traditional boundary-layer flow, and hence will lead to larger overall loading of long structures such as long-span bridges or power transmission cables.

Thunderstorms are responsible for about 1/3 of the extreme gust speeds in the United States (Thom, 1969). In recent studies of extreme wind speeds in the United States, Vickery

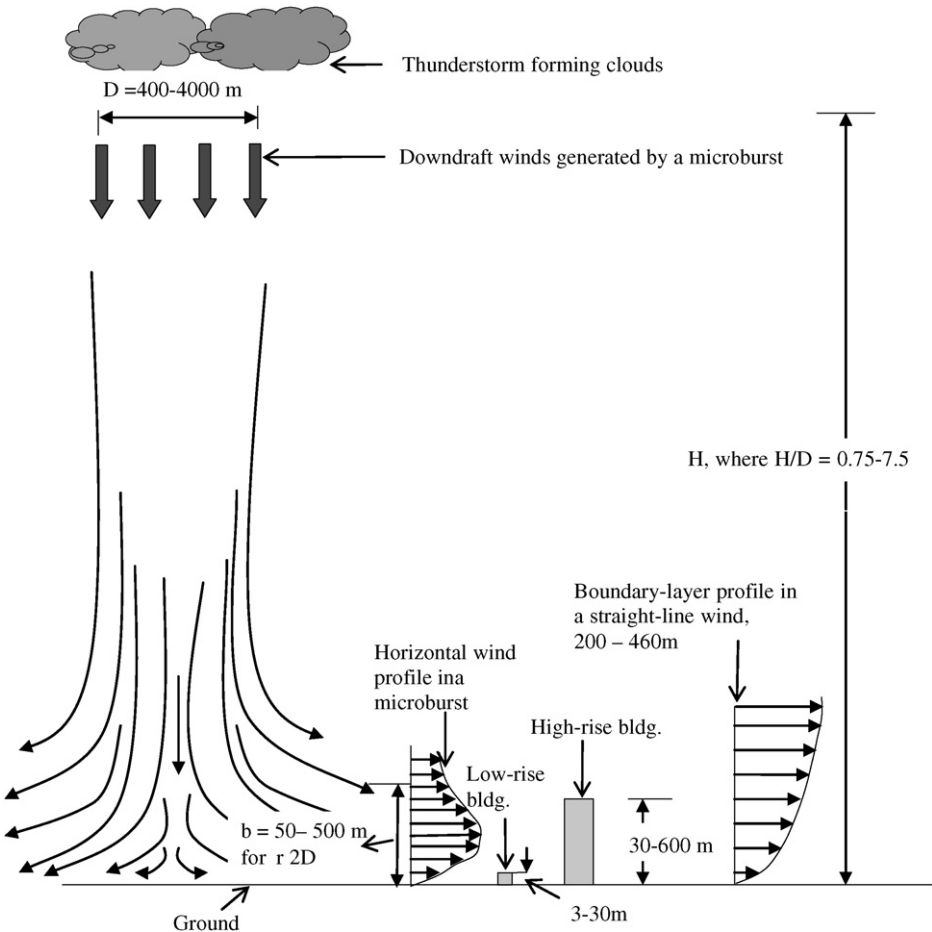


Fig. 1. Typical dimensions of horizontal wind profile in a typical microburst in comparison with straight-line wind profile and building heights.

and Twisdale (1995) found that, outside of hurricane regions, up to 75% of the peak gust wind speeds occurred during thunderstorms.

The current research focuses on simulation of a stationary microburst using both experimental and numerical methods. The microburst is modeled as an impinging jet. Jet impingement is a technique for enhancing heat transfer that is used in a variety of cooling applications from electronics to gas turbines. An intensive parametric study was conducted with different jet exit velocities, height of jet above impingement surface and different nozzle diameters.

Experimentally, the microburst is modeled as a jet coming out of a duct and impinging on a wooden platform placed below. This facility consists of a wooden surface on an adjustable base mounted perpendicular to a round jet nozzle as shown in Fig. 2a. A small centrifugal fan serves as the air supply and is mounted on top of the nozzle. The jet to surface height can be varied from 203 mm (8") to 826 mm (32.5"). The jet was operated at 2 different velocities, 10 and 16 m/s, resulting in a Reynolds number of 1.4×10^5 and 2.2×10^5 , respectively, based on the diameter of the jet exit nozzle and jet exit velocity. Hot-wire anemometry, pressure rakes and particle image velocimetry (PIV) techniques were used for measuring the velocity profiles. Digital pressure transducers were used to measure the pressures on the ground plane. The PIV and pressure measurement setups are shown in Fig. 2a.

Different turbulence models, such as k -epsilon (k - ϵ) and its variants, k -omega (k - ω), and Reynolds stress model (RSM), large eddy simulation (LES) with different boundary conditions and size of computational domain were used to determine the best possible numerical scheme for simulation of the flow-field in a microburst. Commercial CFD software FLUENT (2003, 2005) was used for this purpose.

2. Background and motivation

Microbursts occur in thunderstorms where the weight of the precipitation and the cooling due to microphysical processes act to accelerate the air downwards. Observations suggest that approximately 5% of all thunderstorms produce a microburst which is characterized by a strong localized down-flow and an outburst of strong winds near the surface. Wind shear is the term for conditions when strong winds change direction very quickly. This occurs in a microburst when the strong downdraft is suddenly redirected in a horizontal direction as it hits the ground. This weather phenomenon and resulting wind shear were first identified by Fujita and his co-workers because of the major aircraft disasters they caused.

Fujita (1985) termed microburst as a small downdraft having an outburst of damaging winds with the horizontal extent of the damaging winds being less than 4 km. This definition has been modified by radar meteorologists: they require the peak-to-peak differential Doppler velocity across the divergent center to be greater than 10 m/s and the distance between these peaks be less than 4 km. A microburst becomes a "macroburst" if the horizontal distance between outflow peaks exceeds 4 km. Also, microburst winds descend and spread outward, unlike tornado winds, which converge and rise. After striking the ground, the powerful outward running gust can wreak further havoc along its path. Damage associated with a microburst is often mistaken for the work of a tornado, particularly if the damage is directly under the microburst. However, damage patterns away from the impact area are characteristic of straight-line winds rather than the twisted pattern of tornado damage. Thus, the signature of damage on the ground can be

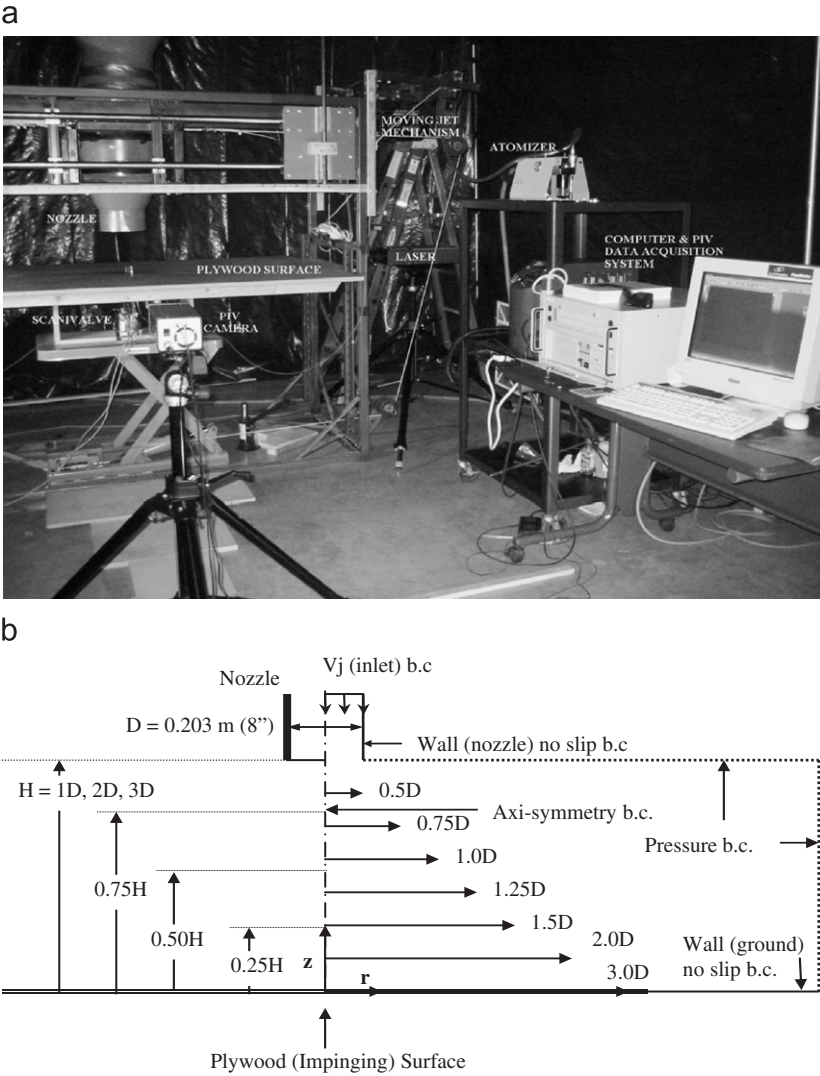


Fig. 2. (a) Experimental setup for PIV and ground pressure measurement. (b) Schematic of jet impingement setup and velocity measurement locations (The boundary conditions used for 2D numerical simulation are shown).

distinguished in a “microburst” compared with a “tornado”. An intense microburst can result in damaging winds with a maximum speed of 270 km/h (170 mph) that often last for less than 5 min.

Selvam and Holmes (1992) undertook numerical modeling of the thunderstorm downdraft phenomenon, and were able to demonstrate reasonable agreement between a numerical model ($k-\epsilon$) and limited full-scale data. Nicholls et al. (1993) performed a 2D LES of a microburst on a building model. Holmes and Oliver (2000) proposed an empirical model of horizontal wind speed and direction generated by a translating downburst.

Holmes (1999), Letchford and Illidge (1999) and Wood and Kwok (1999) undertook physical model studies of a jet impinging on a wall to study topographic effects on velocity profiles in a microburst. Chay and Letchford (2002) and Letchford and Chay (2002) measured velocity profiles on a ground plane in a stationary as well as a moving microburst using an inverted impinging jet. In some of these studies, the comparison of horizontal velocity profiles with those recorded in full scale has shown that impinging jets do a fairly good job in capturing the mean velocities in a steady-state downdraft. Recently, microburst outflow characteristics have been studied by Lin and Savory (2006) and Kim and Hangan (2007). In reality, the microburst is a very short-term phenomenon and usually translates horizontally. A ring vortex has been observed to form in a microburst as the leading edge of the downdraft interacts with the ground and the flow spreads outwards. The horizontally translating ring vortex in a stationary or moving microburst is expected to produce dynamic effects on the velocity profiles. Lundgren et al. (1992), Yao and Lundgren (1996) and Alahyari and Longmire (1995) have conducted several laboratory experiments using fluids of different density in a study of downburst for the aircraft industry. They were able to incorporate the effect of buoyancy in their simulation and produce a ring vortex. Other experimental investigations of jet impingement on flat surfaces were performed by Tsuei (1962), Donaldson and Snedeker (1971), Didden and Ho (1985) and Knowles and Myszko (1998).

The impinging jet flow has been a particularly challenging case for numerical turbulence models. The stagnation region flow is dominated by normal straining of the fluid and many of the widely used models, which have been developed primarily for shear flow boundary layers fail to predict the response of the turbulence to normal straining. Craft et al. (1993) reported predictions of the flow using four different turbulence models using the experimental results of Cooper et al. (1993). They concluded that standard linear eddy viscosity models (EVM) significantly overpredict turbulence energy levels (and thus heat-transfer rates) in the stagnation region, as a result of the linear Boussinesq stress-strain relation misrepresenting the normal stresses and leading to excessive turbulence energy generation rates. They also reported predictions employing stress-transport models. They showed that the linear IP model, when used in conjunction with the wall-reflection terms of Gibson and Launder (1978), gave results that were slightly better than the linear EVM. This failure was due to the form of the wall-reflection terms, which were developed by considering flow parallel to a wall, and actually have the effect of increasing the stress normal to the wall in impinging flow. When these were replaced with the proposed model of Craft et al. (1997), which was designed to predict flows both parallel and normal to a wall, reasonable predictions were obtained. Myszko and Knowles (1995) found out that the standard k - ϵ model failed to accurately predict the thickness of the wall jet. Yamamoto (1997) used a multiple-time-scale RSM to show reasonable agreement with experimental data for mean velocity profiles and growth rates. Dianat et al. (1996) found out that the inclusion of a wall reflection term in Reynolds stress closure model resulted in better predictions of the mean velocities. Recently the v2f model (Behnia et al., 1998) has shown a lot of promise in predicting the behavior of turbulent impinging jets.

The existing literature reveals that impinging jets have been extensively (but not exhaustively) studied in the past, especially for the case of jet impingement in a stagnant environment or in a co-flow configuration. However, a systematic parametric study of the

impinging jet, such as variation of jet diameter, jet exit velocity and height of jet with respect to ground plane using multiple flow-measurement instruments (hotwire, PIV and pressure rake) has not been undertaken in previous laboratory simulations of microburst. An important requirement for employing numerical simulation to gain insight into the flow physics is accurate specification of the appropriate initial, inlet and boundary conditions from physical experiments. In many of the existing numerical studies of microburst simulation there was only a limited set of laboratory data available to the numerical analyst for comparison. Data sets from a parametric study would be very useful in validating CFD results.

The primary objectives of the present work were (a) to perform a systematic parametric study of laboratory simulation of stationary and moving microburst using an impinging jet to determine the flow characteristics; (b) application of CFD to simulate the microburst phenomenon with jet impingement using different turbulence models; (c) comparison of CFD results with those of the experiments to determine the most suitable turbulence model, computational domain and boundary conditions; (d) study of pressure distribution on two cube-shaped buildings in a microburst using both numerical and experimental models to determine the effects of size of the building and its location with respect to the center of the microburst and also to determine the most suitable turbulence model. This paper presents results from the work related to all the above objectives except objective (d) and a moving microburst that will be presented separately.

3. Experimental setup

The experimental setup is shown in Fig. 2. A nozzle diameter (D) of 203 mm (8 in) was used to produce the jet. The distance (H) of the ground plane or impinging platform ($L = 2.44$ m (96 in) and $B = 1.22$ m (48 in) from the nozzle could be varied from a minimum of 203 mm (8 in) to a maximum of 826 mm (32.5 in) using a mechanical jack. A honeycomb and two screens were used to reduce the turbulence of the issuing jet. A 3:1 area contraction was used at the nozzle end to make the velocity of the issuing jet uniform. For the current work, $H = 206$ mm (8.125 in), $H/D = 1.02$; $H = 403$ mm (15.875 in), $H/D = 1.98$; and $H = 587$ mm (23.25 in), $H/D = 2.91$ were used. Henceforth, these three heights are termed as H_1 , H_2 and H_3 , respectively. The H/D ratio for this study was chosen to be within the range of H/D values of a microburst that varies between 0.75 and 7.5. Two jet velocities were used, $V_j = 10$ m/s (32.8 ft/s) and $V_j = 16$ m/s (52.5 ft/s). Henceforth, these two velocities are termed as V_1 and V_2 , respectively. The velocity profiles were measured at different r/D ratios, where r is the radial distance from the centerline of the jet, ranging from 0.5 to 3.0 (Fig. 2b).

3.1. Velocity measurement

A PIV system was used for non-obtrusive velocity measurements. To obtain PIV measurements, the flow was homogeneously seeded with tracer particles (olive oil). In PIV, it is assumed that the particles are small enough to move with the local flow velocity. A plane within the flow is illuminated twice within a short interval of time by a laser sheet. A CCD sensor records on separate frames the light from each pulse scattered by the tracer particles. It is possible to identify the path that a particle has traveled by analyzing one image pair. Knowing the time delay between two pulses, one can

calculate the velocity. The time interval between the two laser shots was adjusted according to the mean flow velocity and the magnification of the camera lens. The PIV system consists of a laser sheet generated by a 120 mJ double pulsed Nd:YAG laser with a maximum repetition rate of 15 Hz per laser head. The CCD camera is a 12-bit camera capable of taking 8 single frames per second, with a minimum interframing time of 300 ns.

Two hundred fifty samples were collected for each run to ensure convergence of statistical quantities, like mean and root mean square (rms) velocity of the flow field. Subsequently, 5–10 runs were conducted at each location to ensure that experimental uncertainties are reduced. An initial interrogation window size of 64×64 pixels and a final initial interrogation window size of 32×32 pixels were used with 50% overlap, resulting in a grid spacing of 16 pixels. An adaptive multipass filter with decreasingly smaller window sizes (2 passes) was used.

Single-probe hot-wire measurements were also performed to measure the axial velocities under the jet as well as the radial velocity profiles at various distances, r , from the centerline of the jet (Fig. 2b). From hot-wire measurements, the turbulence intensity of axial velocities in the jet was found to lie between 1.0% and 2.0%.

A boundary-layer rake consisting of 16 pitot-static tubes was also used to measure the radial velocities at different r/D ratios. The first tube was placed at a height of 2 mm and the last one was at 140 mm from the ground plane with a uniform spacing of 9.2 mm between the tubes. The pressure rake was connected to a Scanivalve pressure transducer (DSA 3217).

3.2. Pressure measurement

Pressures on the ground plane, directly beneath the jet, were measured using 25 taps. One tap was placed directly under the center of the jet and 12 taps were placed on either side of it in a straight line at an interval of 25.4 mm (1"). Two Digital Sensor Arrays (DSA 3217, 16 Channel) Scanivalve pressure transducers were used for measuring both the static and fluctuating surface pressures. The DSA 3217 is a stand-alone temperature compensated electronic pressure scanner with a pressure range of ± 10 in. of H_2O . Each DSA incorporates 16 individual, temperature-compensated, piezoresistive pressure sensors with an A/D converter and a microprocessor to create an intelligent gas pressure measurement system. The maximum data acquisition rate for the scanner was 200 samples/channel/sec. A binary Telnet program, BTEL.EXE, which is a support program for the DSA module along with DSA Link, was used for pressure data acquisition. BTEL can receive and save binary-formatted data from the DSA to a file. It then converts the binary data to ASCII, in a format compatible with spreadsheet programs, like Excel. The data acquisition (using BTEL and DSA Link), was facilitated with a PC running the Windows NT operating system.

4. Numerical CFD modeling

Reynolds averaged Navier–Stokes (RANS) equations are given by

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = 0 \quad (1)$$

and

$$\frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial u_l}{\partial x_l} \right) \right] + \frac{\partial}{\partial x_j}(-\rho \overline{u'_i u'_j}), \quad (2)$$

where the term $-\rho \overline{u'_i u'_j}$ needs to be numerically modeled to close the equation.

The current study incorporates various turbulence models for studying jet impingement flow characteristics with emphasis on the effects of this kind of flow on nearby structures. All the simulations were carried out under incompressibility assumptions. Below is a brief discussion of some of the models, which were used in this research.

4.1. Standard k – ε model

This is still the most frequently used model in general-purpose CFD codes. The model proposed by [Launder and Spalding \(1972\)](#) is used here. The standard model uses the following transport equations for turbulent kinetic energy (k) and its rate of dissipation (ε):

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \varepsilon - Y_M + S_k \quad (3)$$

and

$$\frac{\partial}{\partial t}(\rho \varepsilon) + \frac{\partial}{\partial x_i}(\rho \varepsilon u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + C_{1\varepsilon} \frac{\varepsilon}{k} (G_k + C_{3\varepsilon} G_b) - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} + S_\varepsilon. \quad (4)$$

The above equations contain 5 adjustable constants $C_{1\varepsilon}$, $C_{2\varepsilon}$, C_μ , σ_k and σ_ε which have default values:

$$C_{1\varepsilon} = 1.44 \text{ and } C_{2\varepsilon} = 1.92, \quad C_\mu = 0.09, \quad \sigma_k = 1.00 \text{ and } \sigma_\varepsilon = 1.30.$$

4.2. RNG k – ε model

The RNG k – ε model (renormalization group theory) which is an extension of the standard k – ε model, is also used here as one of the turbulence models. The RNG model differs from the standard k – ε model by inclusion of an extra term R in the right-hand side of the ε equation. The extra term R depends on the rate of strain, as given by

$$R = \frac{\rho C_\mu \eta^3 (1 - (\eta/\eta_0))}{(1 + \beta \eta^3)} \left(\frac{\varepsilon^2}{k} \right), \quad (5)$$

where $\eta = Sk/\varepsilon$. The model constants for RNG k – ε model ([FLUENT, 2003](#)) are given by:

$$C_{1\varepsilon} = 1.42, \quad C_{2\varepsilon} = 1.68, \quad C_\mu = 0.0845, \quad \sigma_k = 1.0, \\ \sigma_\varepsilon = 1.3, \quad \eta_0 = 4.38 \text{ and } b = 0.012.$$

4.3. Realizable k – ε model

Here the transport equations are given by

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \varepsilon - Y_M + S_k \quad (6)$$

and

$$\frac{\partial}{\partial t}(\rho\varepsilon) + \frac{\partial}{\partial x_j}(\rho\varepsilon u_j) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + \rho C_1 S_\varepsilon - \rho C_2 \frac{\varepsilon^2}{k + \sqrt{\nu \varepsilon}} + C_{1\varepsilon} \frac{\varepsilon}{k} C_{3\varepsilon} G_b + S_\varepsilon, \quad (7)$$

where

$$C_1 = \max \left[0.43, \frac{\eta}{\eta + 5} \right], \quad \text{and} \quad \eta = S \frac{k}{\varepsilon}.$$

The model constants for the realizable k – ε model are given by

$C_{1\varepsilon} = 1.44$, $C_2 = 1.68$, $\sigma_k = 1.0$ and $\sigma_\varepsilon = 1.2$. C_μ is no longer a constant in this model. More details about this model can be found in [Shih et al. \(1995\)](#).

4.4. Shear–stress transport (SST) k – ω model

This is a variation of the standard k – ω model proposed by [Wilcox \(1998\)](#). In this model, in addition to the standard k equation, another equation for length (ω) is solved. This quantity (ω) is often called specific dissipation and it is defined as $\omega \propto \varepsilon/k$. The modeled k and ω equations are given by ([FLUENT, 2003](#))

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\Gamma_k \frac{\partial k}{\partial x_j} \right] + G_k - Y_k + S_k \quad (8)$$

and

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_i}(\rho \omega u_i) = \frac{\partial}{\partial x_j} \left[\Gamma_\omega \frac{\partial \omega}{\partial x_j} \right] + G_\omega - Y_\omega + D_\omega + S_\omega. \quad (9)$$

In the SST k – ω model, the definition of turbulent viscosity is modified to account for the transport of the principal turbulent shear stress and this feature enables this model to perform better than the standard k – ε and standard k – ω model. There is also a cross-diffusion term D_ω added to the ω equation.

4.5. Reynolds stress model (RSM)

The transport equations for transport of Reynolds stresses, $\overline{\rho u'_i u'_j}$ are written as follows:

$$\begin{aligned} \frac{\partial}{\partial t}(\overline{\rho u'_i u'_j}) + \frac{\partial}{\partial x_k}(\overline{\rho u_k u'_i u'_j}) = & - \frac{\partial}{\partial x_k} [\overline{\rho u'_i u'_j u'_k}] + \overline{\rho(\delta_{kj} u'_i + \delta_{ik} u'_j)}] + \frac{\partial}{\partial x_k} \left[\mu \frac{\partial}{\partial x_k} (\overline{u'_i u'_j}) \right] \\ & - \rho \left(\overline{u'_i u'_k} \frac{\partial u_j}{\partial x_k} + \overline{u'_j u'_k} \frac{\partial u_i}{\partial x_k} \right) - \rho \beta (g_i \overline{u'_j \theta} + g_j \overline{u'_i \theta}) + p \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right) - 2\mu \overline{\frac{\partial u'_i}{\partial x_k} \frac{\partial u'_j}{\partial x_k}} \\ & - 2\rho \Omega_k (\overline{u'_j u'_m} \varepsilon_{ikm} + \overline{u'_i u'_m} \varepsilon_{jkm}) + S_{\text{user}}. \end{aligned} \quad (10)$$

More details about this model and definition of the individual terms can be found in [FLUENT \(2003\)](#); [Gibson and Launder \(1978\)](#) and [Launder et al. \(1975\)](#).

4.6. Large eddy simulation (LES)

In LES, the governing equations are derived by filtering the time dependent Navier–Stokes equations. The eddies whose scales are smaller than the filter width or grid spacing are filtered out. The subgrid-scale stresses resulting from the previous filtering operation require modeling. The Boussinesq hypothesis (FLUENT, 2005) is employed and the subgrid-scale turbulent stresses are computed from

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2\mu_t\bar{S}_{ij}, \quad (11)$$

where μ_t is the subgrid-scale turbulent viscosity, and $\bar{S}_{ij}$ is the rate-of-strain tensor for the resolved scale defined by

$$\bar{S}_{ij} \equiv \frac{1}{2} \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right). \quad (12)$$

For the calculation of μ_t , the dynamic Smagorinsky–Lilly model (FLUENT, 2005) was used. The eddy viscosity μ_t is modeled as

$$\mu_t = \rho L_s^2 |\bar{S}|, \quad (13)$$

where L_s is the mixing length for subgrid scales and

$$|\bar{S}| \equiv \sqrt{2\bar{S}_{ij}\bar{S}_{ij}}. \quad (14)$$

L_s is calculated using

$$L_s = \min(\kappa d, C_s V^{1/3}), \quad (15)$$

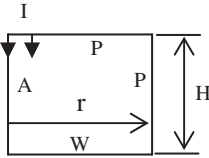
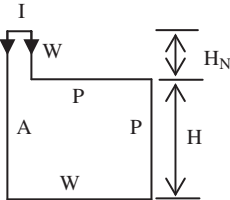
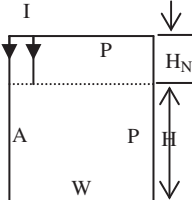
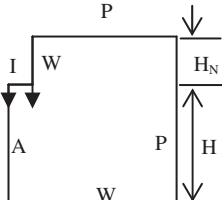
where κ is the von Karman constant, d is the distance to the closest wall, C_s is the Smagorinsky constant, and V is the volume of the computational cell. C_s is computed dynamically based on the information provided by the resolved scales of motion and varies in time and space over a wide range.

4.7. Solution parameters

For all the numerical simulations, the density and viscosity of air were taken as 1.225 kg/m³ and 1.7894 E–05 N s/m², respectively. The inlet values of turbulence intensities were substituted from the experimental data. For all the simulations using RANS models, the QUICK scheme was used for the convective fluxes as this reduces the numerical diffusion, and a 2nd order scheme was used for k and ε terms. For LES, a central difference scheme was used. The SIMPLE algorithm was used for the pressure correction for steady-state simulations, whereas for LES the PISO algorithm was used.

Initially, the numerical simulation was carried out using an axisymmetric jet with various domain dimensions (r , H , H_N), as listed in Table 1, using the k – ε model. Axisymmetric boundary condition (b.c.) was used along the centerline of the jet. It was observed that the difference in the radial velocities did not vary by more than 2.8% for these cases (C1–C6). The axial velocity profiles under the jet for all these simulations were also in close agreement with each other. Ultimately, it was decided to adopt case C2 for our analysis, based on considerations of optimum domain size as well as of computation time.

Table 1
Shapes and sizes of domains tested

Case C1	Cases C2 and C3		Case C4 and C5	Cases C6
				
$H = 2D$	$H = 2D$		$H = 2D, H_N = D$	$H = 2D, H_N = D$
C1	C2	C3	C4	C5
$r = 5D$	(a) $r = 5D$	(a) $r = 5D$	(a) $r = 5D$	(b) $r = 10D$
	(b) $H_N = D$	(b) $H_N = D/4$		$r = 5D$

P = pressure b.c., W = wall b.c., A = axisymmetry b.c., I = inlet b.c. (uniform jet), D = jet diameter, H = jet height, r = radial distance from jet centerline, H_N = height of jet nozzle.

The comparison of normalized radial velocity profiles at r/D of 0.5, 1, 2 and 3 that were obtained with $k-\varepsilon$ model for different boundary conditions, cases C1–C6 as described in Table 1, is shown in Fig. 3. It shows good agreement proving that these boundary conditions have little influence on the flow profiles.

Fig. 4a shows a typical grid used for LES. To ensure grid independency of the LES simulations, 3 different grid sizes were tested; namely, coarse (1.4 million cells), medium (2 million cells) and fine (2.8 million cells). It can be seen in Fig. 4b, that the results of the medium and fine grids do not differ much. As such all the LES results presented henceforth are based on the medium grid only.

5. Results and discussion

The ground pressure coefficients are shown in Fig. 5, where $C_p = (P - P_s)/0.5\rho V_j^2$. The ground pressures corresponding to the four jet heights (H_1 , H_2 , H_3 and H_4) and two velocities (V_1 and V_2) match exactly (as seen in Fig. 5a) showing that there is no effect of the jet height and jet velocity on ground pressures for the range of parameters chosen. It can be seen that the numerical values of pressure coefficients calculated using the different turbulence models match very well with that obtained from the experiments (Fig. 5b). The uncertainty in the experimental pressure values was estimated as less than 10%.

Flow visualization using PIV is shown in Fig. 6 where it can be observed that jet axial velocity is uniform across the width of the nozzle near its exit and at mid-height from the ground plane but starts to show the effects of the ground plane as the jet nears the impingement surface. The jet is observed to expand slightly as it descends.

The axial velocity profiles along the centerline of the jet at various heights from the ground plane are shown in Fig. 7. The comparison in Fig. 7a clearly shows that the jet expands along its edges as it descends and the velocity at the center of the jet decreases by almost 30% as it approaches the ground plane. It also shows the consistency of the profiles between two different velocities V_1 and V_2 of the jet (Fig. 7a). There is very good agreement between the experimental measurements and numerical (RSM) data as seen in

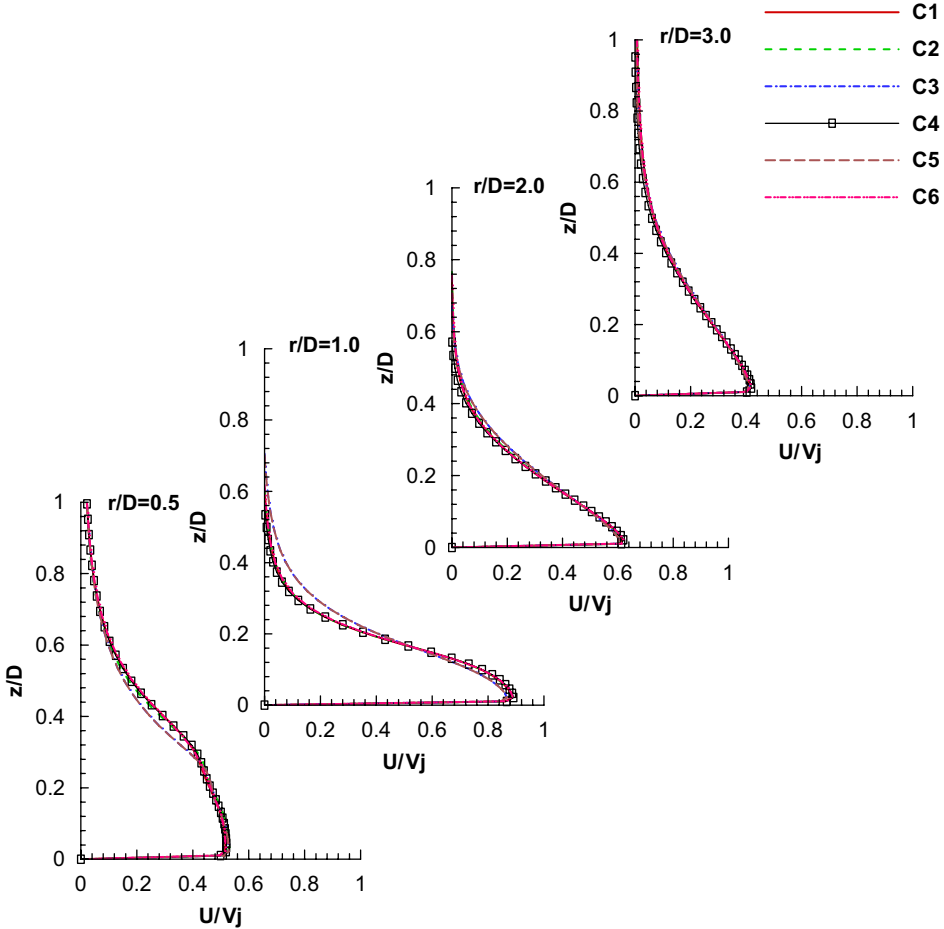


Fig. 3. Comparison of radial velocities with different numerical boundary conditions.

Fig. 7b. Axial velocity profiles from other numerical methods also match very well with the experimental ones but are not shown here. The PIV and hot-wire measurements differ near the edges of the jet. The PIV profiles consistently show a narrower jet at different heights. This could be due to reduced concentration of seeding particles along the jet boundary; a common problem in the use of PIV to visualize jet flows.

A comparison of the normalized axial velocities along the centerline of the jet obtained with PIV and all the numerical models shows (Fig. 8) close agreement between the experimental and all the numerical models. This plot also shows how the jet velocity along the centerline decreases as the jet approaches the ground.

The normalized profiles of horizontal velocity (U) at different r/D values are plotted in Fig. 9a and compared with the empirical profiles of Rajaratnam (1976) and Wood et al. (2001). The general form of the empirical profile is given by

$$\frac{U}{U_m} = C_1 \left(\frac{z}{b} \right)^n \left[1 - \operatorname{erf} \left(C_2 \frac{z}{b} \right) \right], \quad (16)$$

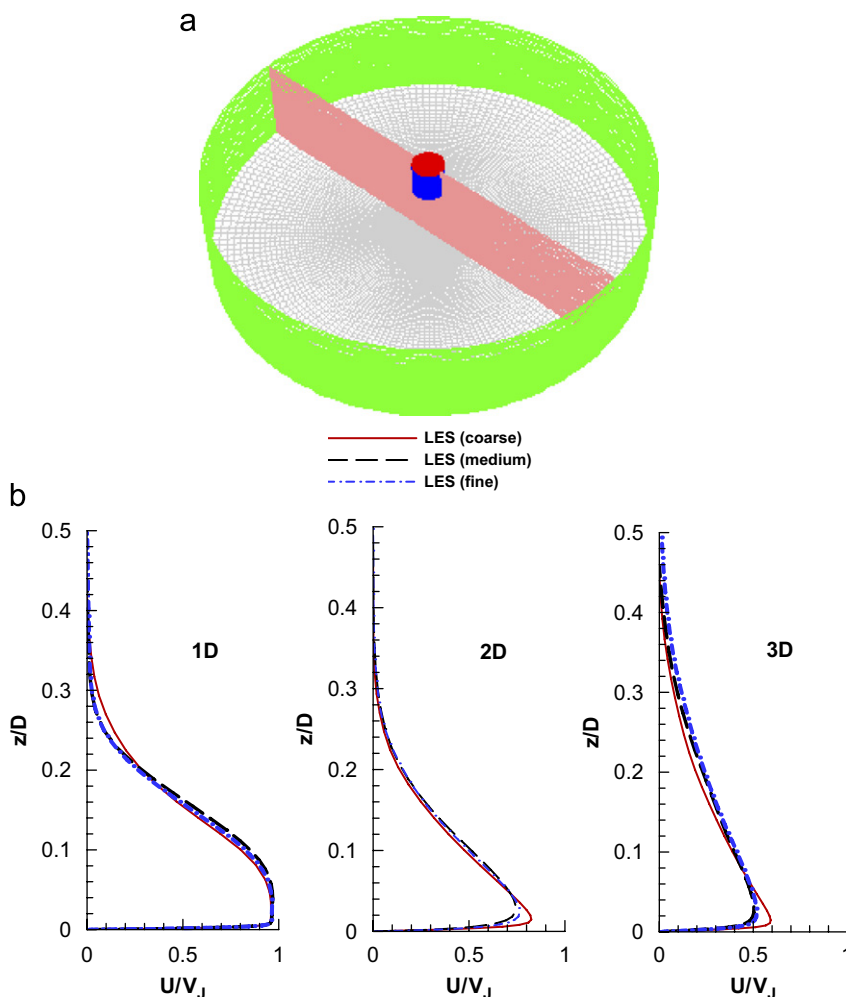


Fig. 4. (a) Grid for LES simulation. (b) Grid independence study in LES at $r = 1D, 2D$ and $3D$.

where $C_1 = 1.48$, $C_2 = 0.68$ and $n = 1/7$ (Rajaratnam, 1976); $C_1 = 1.55$, $C_2 = 0.70$ and $n = 1/6$ (Wood et al., 2001); U_m = maximum horizontal velocity at a specific r/D ; b = elevation from the ground at which $U = U_m/2$ and $\text{erf}()$ is error function.

The current experimental data from PIV and pressure rake at several r/D locations (Fig. 2b) were used to generate Eqs. (16)–(18). The uncertainty in the velocity measurements depends on the flow region but generally it was estimated to be within $\pm 10\%$.

The best curve-fit of these data (for $r/D = 0.75, 1, 1.25, 1.5, 2.0, 2.5$ and 3.0) is obtained with $C_1 = 1.52$, $C_2 = 0.68$ and $n = 1/6.5$ in Eq. (16). This empirical profile is plotted in Fig. 9a and was found to agree very well with those of Rajaratnam (1976) and Wood et al. (2001). Also plotted in the same figure are the experimental results of Cooper et al. (1993) and Chay and Letchford (2002). The range of r/D for which the proposed empirical profiles are valid is $0.75 \leq r/D \leq 3.00$. In the figure, the normalized experimental data for $r/D = 1, 2$ and 3 only are shown for sake of clarity.

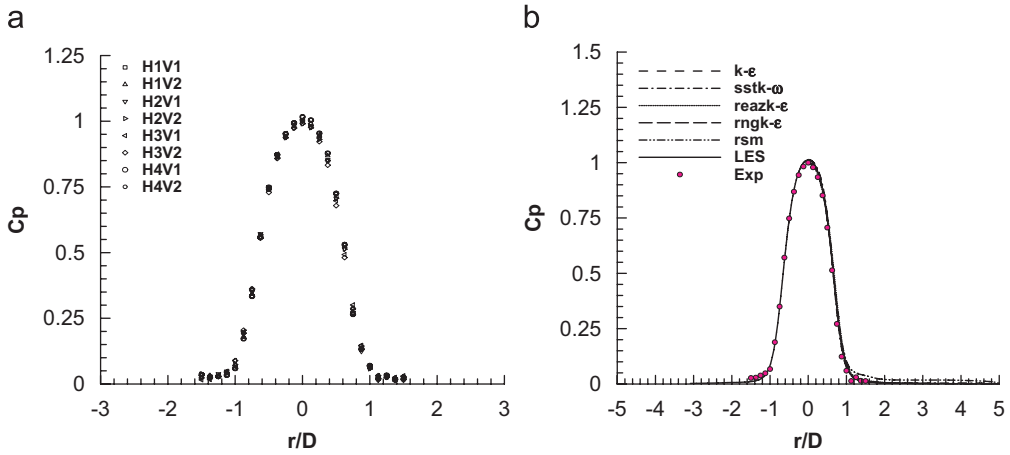


Fig. 5. (a) Plot of pressure coefficient on impingement ground plane (experimental). (b) Comparison of experimental and numerical simulation for $V1$.

The field observation data from Project NIMROD (Fujita, 1981) is also shown in the same plot. The observed values agree closely at higher elevations up to z/b of 1 compared with those at lower ones ($z/b < 0.2$). This could be due to the fact that NIMROD Doppler radar data is prone to errors at lower elevations because of obstructions near ground level. The mismatch between Doppler data and current experimental data at elevations corresponding to $z/b > 1$ could be attributed to higher scatter in the experimental data due to measurement errors of relatively smaller radial velocities ($U/U_m < 0.45$) at these elevations.

The numerical results for normalized radial velocity profiles of the jet are compared with the empirical profile (Eq. (16)) obtained from experimental data and found to agree well, as shown in Fig. 9b.

The normalized curve for U_m that fits the experimental data (PIV and rake), as plotted in Fig. 10a, is given by

$$\frac{U_m}{V_j} = \exp \left[1.905 - \frac{1.858}{r/D} - 1.949 \ln \left(\frac{r}{D} \right) \right]. \quad (17)$$

The maximum radial velocity (U_m) is almost equal in magnitude to the jet velocity (V_j) up to r/D of 1 and then decreases to $0.425V_j$ at r/D of 3.

The curve for the prediction of boundary-layer development of the jet that fits the experimental data, as shown in Fig. 11a, is given by

$$\frac{b}{D} = -0.0362 + 0.0818 \frac{r}{D} + 1.0062 \exp \left[-2.472 \left(\frac{r}{D} \right) \right]. \quad (18)$$

The numerical results for normalized curve for U_m and the boundary layer development of the jet do not match that well with the experimental data as shown in Figs. 10b and 11b, respectively. From an assessment of various numerical turbulence models, it is seen that the realizable $k-\epsilon$ model gives results that are closest to the experimental values for the entire range of r/D , whereas RSM model gives acceptable results up to r/D of 2.

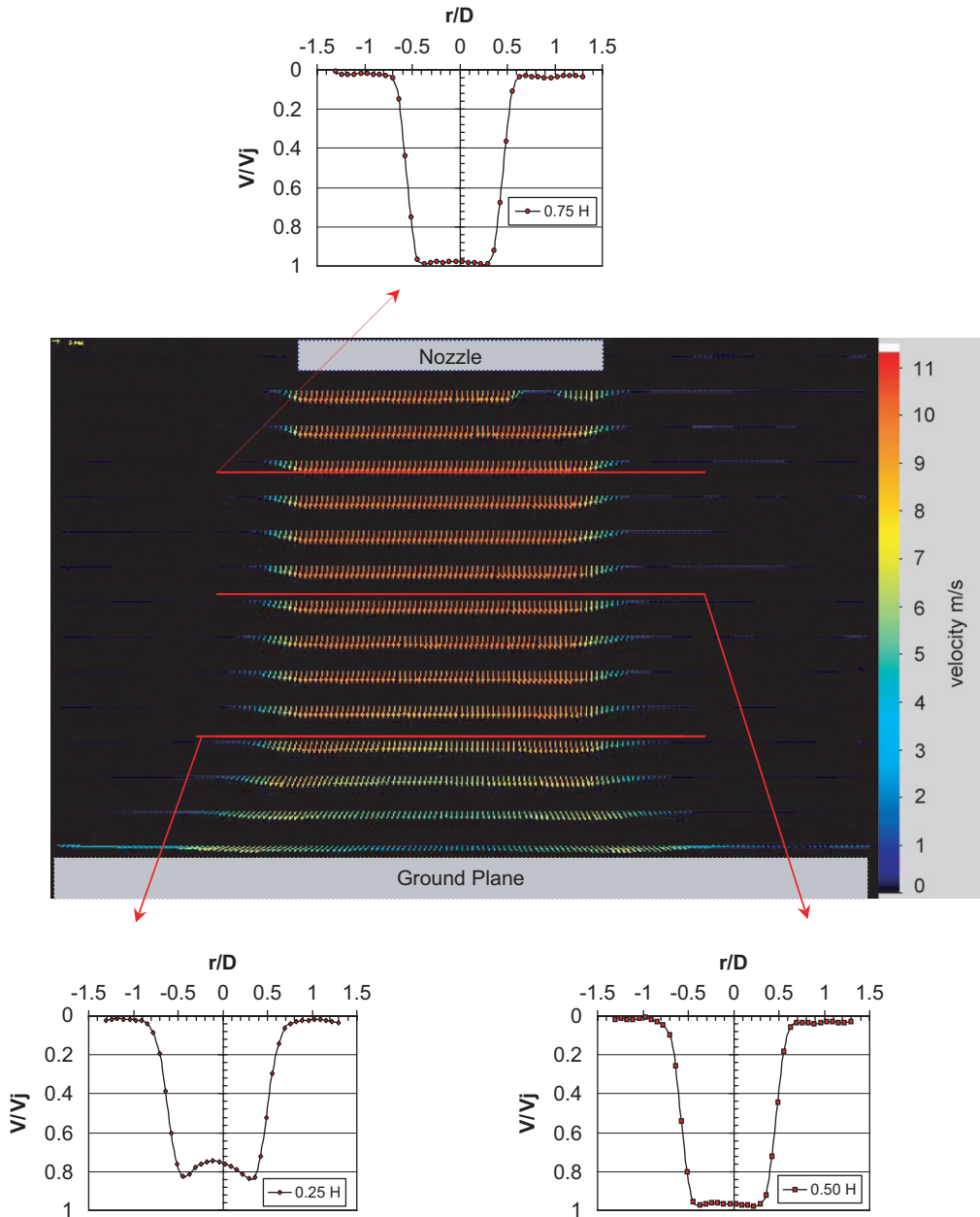


Fig. 6. Jet impingement visualization from PIV.

The standard $k-\varepsilon$, RNG $k-\varepsilon$ and SST $k-\omega$ models produce results that deviate the most from the experimental values.

For understanding the effect of nozzle diameter on the radial velocities, the experimental data of Tsuei (1962), Cooper et al. (1993) and Chay and Letchford (2002) were compared

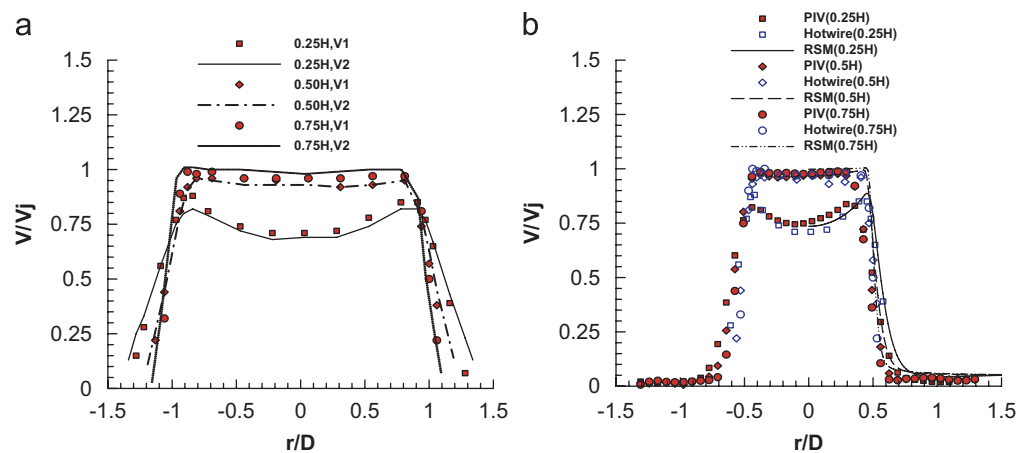


Fig. 7. (a) Comparison of normalized jet velocities (hot-wire data) at different normalized heights. (b) Comparison of experimental (PIV, hot wire) and numerical (RSM) normalized vertical velocity profiles at $H = 2D$ and $V = V_1$.

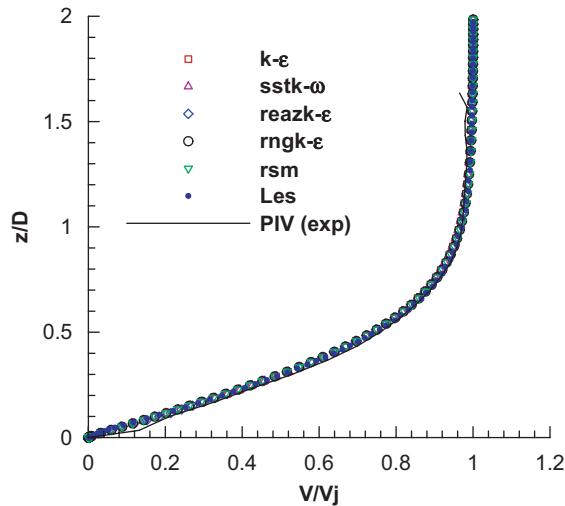


Fig. 8. Comparison of normalized vertical velocities along the centerline of the jet (experimental and numerical).

with the empirical profiles for U_m and b obtained for $D = 203$ mm (8 in) nozzle (Fig. 12). There is generally very good agreement between the proposed profiles and those of Tsuei (1962), Cooper et al. (1993) and Chay and Letchford (2002). It is therefore concluded, based on this limited comparison, that the normalized profiles are not a function of jet diameter. This has a positive implication on the geometric scaling-up of the experimental results to the field.

Table 2 shows the values of U_m/V_j and b/r corresponding to different jet heights, jet exit velocities and location x from the jet centerline. It can be seen that these values correspond

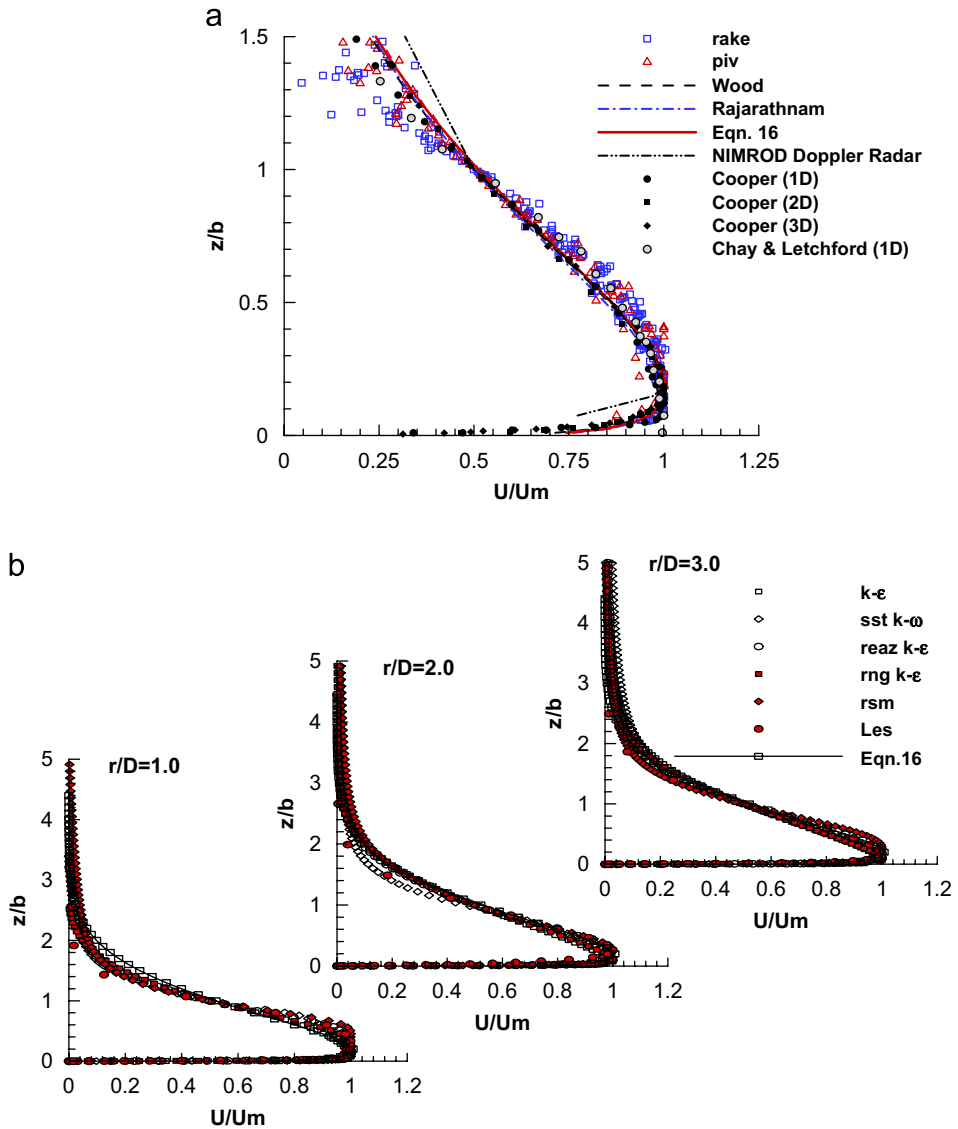


Fig. 9. (a) Comparison of normalized radial velocity profiles (experimental with pressure rake, NIMROD field data and empirical profiles as given by Eq. (16). (b) Comparison of numerical simulation data for $H/2$ and $V/1$ with present empirical profile as given by Eq. (16).

fairly well with the values given in Blevins (1984) for impinging radial jet ($H/D > 1$) with $r/D \geq 1$, but not so well for $r/D < 1$ as originally specified in Blevins (1984).

For understanding the second order statistics, a comparison of the radial turbulence intensity profiles was made. Fig. 13 shows the results of the comparison of the data obtained using PIV and LES. It is seen that there is reasonable agreement at most locations except at $r/D = 0.75$ and 1.0 , which is the transition region.

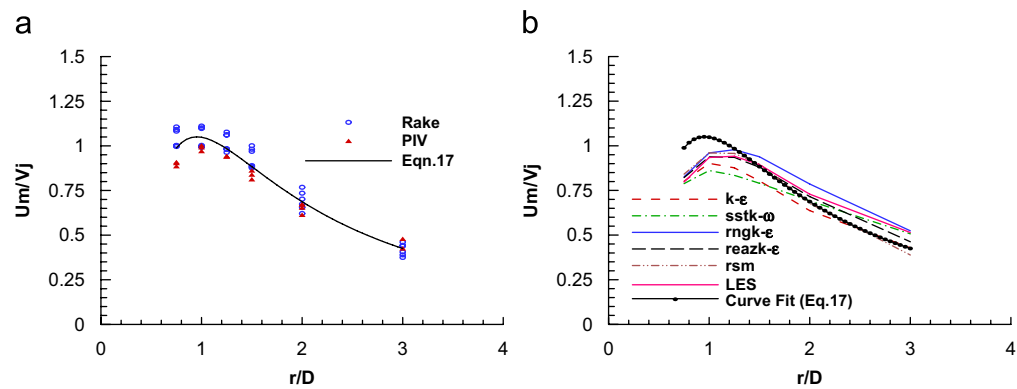


Fig. 10. (a) Plot of U_m/V_j as a function of r/D , (b) Comparison of numerical simulation data for $H2$ and $V1$ with present empirical profile as given by Eq. (17).

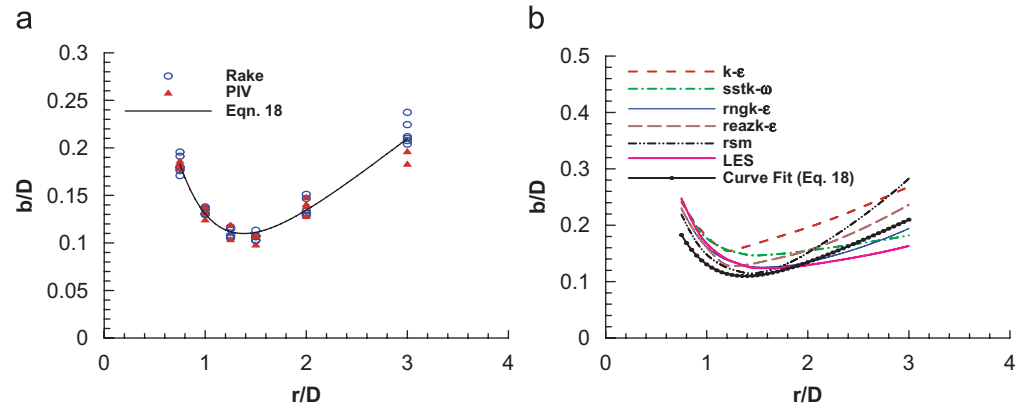


Fig. 11. (a) Plot of boundary-layer growth, (b) Comparison of numerical simulation data for $H2$ and $V1$ with present empirical profile as given by Eq. (18).

6. Summary and conclusions

A laboratory study of a stationary microburst using an impinging jet was conducted to systematically evaluate the effect of different parameters on the velocity profiles and pressure distribution on the ground. Numerical simulations were carried out with uniform jet profiles and proper boundary conditions and extent of the numerical domain were identified. Different turbulence models were applied to determine their applicability to this type of flow. Comparison of numerical and experimental results was made. The empirical equations derived here from the experimental data have potential application in the design of civil engineering structures for winds generated by a microburst.

The jet expands and the velocity along the centerline of the jet decreases as it gets closer to the ground. There is good agreement between axial velocity profiles of the jet obtained from experiments (hot wire) with those obtained with various numerical turbulence models. The PIV was unable to capture the entire width of the jet because of possible lack

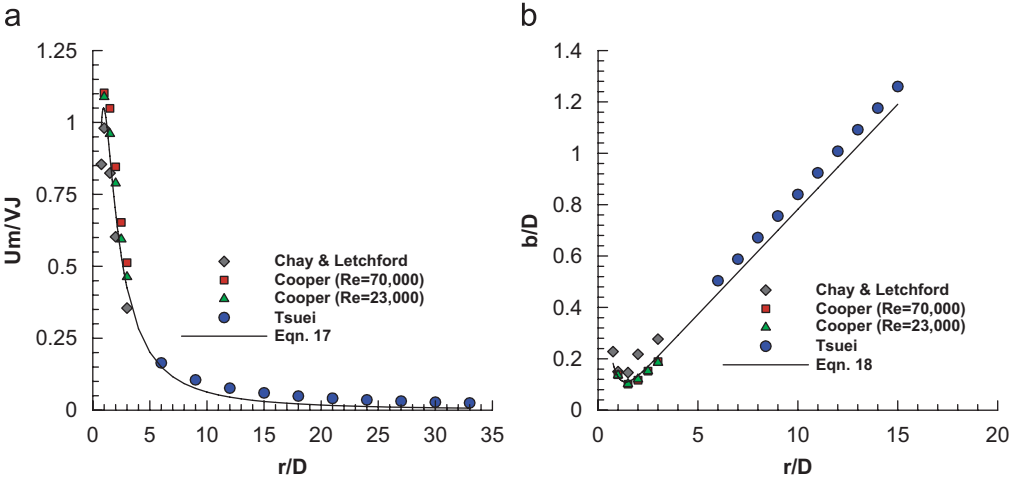


Fig. 12. Comparison of data obtained by Cooper et al. (1993) and Tsuei (1962) with empirical profiles for $D = 203\text{ mm}$ (8") (a) U_m/V_j vs. r/D and (b) b/D vs. r/D .

Table 2
Comparison of current experimental data with Blevins (1984)

H (m) $r/H \geq 0.25$	V_j (m/s)	r	U_m/V_j Eq. (17)	U_m/V_j Blevins (1984) ^a $r > 0.22D$	b/r Eq. (18)	b/r Blevins(1984) ^a $r > 0.22D$
$H1-H3$	$V1$ and $V2$	$0.75D$	0.99	1.40	0.244	0.087
$H1-H3$	$V1$ and $V2$	$1D$	1.05	1.05	0.131	0.087
$H1-H3$	$V1$ and $V2$	$2D$	0.69	0.53	0.068	0.087
$H1-H3$	$V1$ and $V2$	$3D$	0.43	0.35	0.070	0.087

^aUncertainty in these values is 20–30%.

of seeding particles along the jet boundary that can be possibly corrected in the future if the ambient air surrounding the jet is also seeded.

The horizontal velocity profiles from experiments agree well with the empirical profiles of Rajaratnam (1976) and Wood et al. (2001), and a new empirical profile equation is proposed. The normalized horizontal velocity profiles computed numerically with various models match closely with the experimental results for different r/D values. New empirical equations are proposed for predicting the boundary layer growth and the maximum radial velocity as a function of distance from the center of the jet. The validity of these equations were checked with the experimental data of other researchers and was found to agree well. All the numerical models capture the trend of the boundary layer growth and decay of maximum radial velocities with increasing radial distance, but the predicted values from LES, realizable $k-\epsilon$ and RSM models match the experimental data better than the other

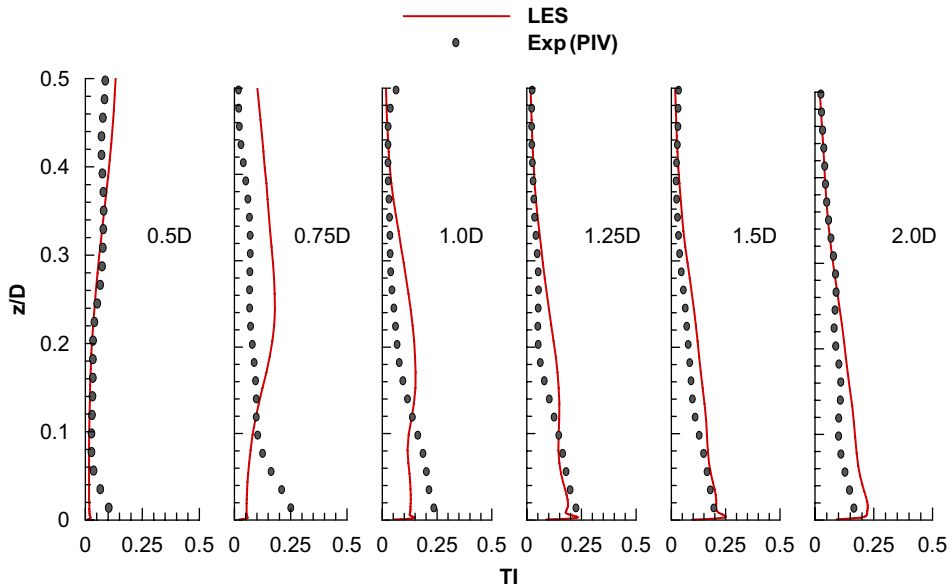


Fig. 13. Radial turbulence intensity profiles.

three models. Both ground pressure coefficients and axial velocity profiles under the jet computed numerically with different turbulence models agree very well with experimental results.

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